I performed an analysis on variants of 4 different proteins using a few different machine learning models to see how well they would perform in identifying gain of function versus loss of function variants with particular emphasis on the gain of function variants.

I am uploading a set of files generated by this analysis of variants. Variants for 4 different proteins were analyzed, MC4R, HXK4, PTEN, SRC. The workflow involved running machine learning models for each protein separately. They were trained using MAVE datasets for each protein. The first model didn't involve training per se. It involved computing the log likelihood ratio of each variant and using that as the score. The second model called "RF AL" involved generating embeddings from an ESM2 model and feeding them to a Random Forest. The training was done using active learning. The third model was the same as the "RF AL" model except that it was trained using a monte carlo method using 20% of the dataset for training. We call this "RF ALL PRED".

The performance of each model on each protein was evaluated and analysis data files were generated. Here are the data files for each analysis.

Filename: protein_landscape_data.csv.

This file contains the following columns: mutation_position (residue position):

assay_score (mave assay score) ,

prediction_score,

count (number of variants at this position) ,

gof_assay_threshold (minimum assay score to be considred gain of function),
lof_assay_threshold (maximum assay score to be considred loss of function),

assay_source(protein),

model(The part after the slash indicates the model used for training, LogLikehood indicates there was no training and log likehood ratios were computed)

The combination of the following columns make a row unique: assay_source, mutation_position, model.

In the case where LogLikehood is the model the prediction_score column is actually the z value of the esm2 log likehood ratio and the assay_score column is the z value of the mave assay score value.

Filename: protein_landscape_domains.csv.

This file contains the domains of the proteins with the following columns:

Name(domain name)

Start(starting position)

End(ending position)

Assay_source(protein)

The combination of the following columns make a row unique: assay_source, domain.

Filename: mse_by_domain.csv.

This file contains the mean squared error by domain for each protein for the "RF AL" model mentioned above. It contains the following columns:

Domain(domain name)

domain_start(start position)

domain_end(end position)

assay_source(protein)

metric(mean squared error)

model(constant value of RF AL)

The combination of the following columns make a row unique: assay_source, model, domain.

Filename: auc_by_domain.csv

This file contains the auc by domain for each protein. The variants are labelled as either GOF or LOF based upon strict criteria found in the papers where the MAVE data is reported. Variants that did not meet either of these strict criteria were not included in the analysis.

Domain(domain name)

domain_start(start position)

domain_end(end position)

optimal_youden_index

assay_source(protein)

metric(ROC AUC for the domain)

model(RF AL or LLR(log likehood ratio))

The combination of the following columns make a row unique: assay_source, model, domain.

Filename: auc_by_label_method_by_domain.csv

This file contains the auc by domain for each protein using labels that were generated using different methods. The model used was the the "RF AL" model described above. The methods used to generate labels are the following:

Label Method: GOF/LOF: 1 means GOF, 0 means LOF. These labels are assigned according to the methodology used in original dataset papers or using a fairly strict classification scheme where labels were assigned only for variants that were clear GOF or LOF. This left many variants unlabelled and unused in the analysis.

Label Method: GOF_10%: Variants with top 10% highest scores were assigned 1 and everything else 0.

Label Method: GOF_20%: Variants with top 20% highest scores were assigned 1 and everything else 0.

Label Method: LOF_10%: Variants with the 10% lowest scores were assigned 0 and everything else 1.

Label Method: LOF_20%: Variants with the 20% lowest scores were assigned 0 and everything else 1.

The columns are:

Domain(domain name)

domain_start(start position)

domain_end(end position)

optimal_youden_index

assay_source(protein)

metric(ROC AUC for the domain for the label method in model column)

model(The label method. They were described above)

The combination of the following columns make a row unique: assay_source, model, domain.

Filename: auc_by_label_method_by_round.csv

This file contains the auc by active learning round for each protein using labels that were generated using different methods. The model used was the the "RF AL" model described above. The methods used to generate labels are the following:

Label Method: GOF/LOF: 1 means GOF, 0 means LOF. These labels are assigned according to the methodology used in original dataset papers or using a fairly strict classification scheme where labels were assigned only for variants that were clear GOF or LOF. This left many variants unlabelled and unused in the analysis.

Label Method: GOF_10%: Variants with top 10% highest scores were assigned 1 and everything else 0.

Label Method: GOF_20%: Variants with top 20% highest scores were assigned 1 and everything else 0.

Label Method: LOF_10%: Variants with the 10% lowest scores were assigned 0 and everything else 1.

Label Method: LOF_20%: Variants with the 20% lowest scores were assigned 0 and everything else 1.

The columns are:

Round_num(active learning round number)

Auc(roc auc for that round)

optimal_youden_index

assay_source(protein)

model(The label method. They were described above) llr_auc(The auc computed when using the log likelihood ratio as the predicted score. It will be constant across rounds)

num_positive – Number of positive variants

num_negative – Number of negative variants

The combination of the following columns make a row unique: round_num, assay_source, model.

Generate a manuscript using the data files for this analysis that would be suitable for submission to publication in a journal. Include a background section, methods section, results, and discussion section, and reference section. Include tables and figures.